Since $\Delta \neq 0$, we get $x_1 = \frac{\Delta_1}{\Delta}$.

Similarly, we get $x_2 = \frac{\Delta_2}{\Delta}, x_3 = \frac{\Delta_3}{\Delta}$, where $\Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$.

Thus, we have the Cramer's rule $x_1 = \frac{\Delta_1}{\Delta}, x_2 = \frac{\Delta_2}{\Delta}, x_3 = \frac{\Delta_3}{\Delta}$,

where $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \Delta_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$

Note

Replacing the first column elements a_{11}, a_{21}, a_{31} of Δ with b_1, b_2, b_3 respectively, we get Δ_1.

Replacing the second column elements a_{12}, a_{22}, a_{32} of Δ with b_1, b_2, b_3 respectively, we get Δ_2.

Replacing the third column elements a_{13}, a_{23}, a_{33} of Δ with b_1, b_2, b_3 respectively, we get Δ_3.

If $\Delta = 0$, Cramer's rule cannot be applied.

Example 1.25

Solve, by Cramer's rule, the system of equations

$x_1 - x_2 = 3, 2x_1 + 3x_2 + 4x_3 = 17, x_2 + 2x_3 = 7.$

Solution

First we evaluate the determinants

$$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{vmatrix} = 6 \neq 0, \Delta_1 = \begin{vmatrix} 3 & -1 & 0 \\ 17 & 3 & 4 \\ 7 & 1 & 2 \end{vmatrix} = 12, \Delta_2 = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 17 & 4 \\ 0 & 7 & 2 \end{vmatrix} = -6, \Delta_3 = \begin{vmatrix} 1 & -1 & 3 \\ 2 & 3 & 17 \\ 0 & 1 & 7 \end{vmatrix} = 24.$$

By Cramer's rule, we get $x_1 = \frac{\Delta_1}{\Delta} = \frac{12}{6} = 2, x_2 = \frac{\Delta_2}{\Delta} = \frac{-6}{6} = -1, x_3 = \frac{24}{6} = 4.$

So, the solution is $(x_1 = 2, x_2 = -1, x_3 = 4)$.

Example 1.26



In a T20 match, a team needed just 6 runs to win with 1 ball left to go in the last over. The last ball was bowled and the batsman at the crease hit it high up. The ball traversed along a path in a vertical plane and the equation of the path is $y = ax^2 + bx + c$ with respect to a xy-coordinate system in the vertical plane and the ball traversed through the points $(10,8), (20,16), (40,22)$, can you conclude that the team won the match?

Justify your answer. (All distances are measured in metres and the meeting point of the plane of the path with the farthest boundary line is $(70,0)$.)

Solution

The path $y = ax^2 + bx + c$ passes through the points $(10,8), (20,16), (40,22)$. So, we get the system of equations $100a + 10b + c = 8, 400a + 20b + c = 16, 1600a + 40b + c = 22$. To apply Cramer's rule, we find

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